Coding sessions (version 1) — annotated

Black, left: the original document, unchanged. **Red, right: review notes.** ✓ = checked numerically (Python/NumPy, RK4, open chains, centre of mass in unit cells of 2 sites). Convention used in all checks: ħ = 1, i dψ/dt = Hψ, $H_{n,n+1}$ = +J, A = odd sites, J_1 inside the unit cell (as in the chain figure).

Coding sessions: Fridays, *time tbd* (e.g. 10:15 - 11:45 or 14:00-15:30) in 56-368

- Building simple Hamiltonians
- Numerically calculating the eigenvalues and eigenstates of a Hamiltonian
  - In real-space
  - In k-space
- The SSH model
  - Defects
  - Symmetry breaking
- Time-evolution (Runge-Kutta)
- Winding number and Zak phase
- Time-modulation
- Beam propagation simulation
- Mach-Zehnder Interferometers and the Bloch sphere
- Coding with AI agents

Tasks:

General

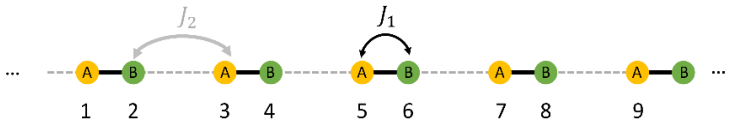
- **Topics vs. tasks:** every topic now has a task, but 11 and 12 are only titles (see notes there). The *winding number* has no task yet; add it to task 8.
- **Missing conventions box.** State once at the top: ħ = 1; i dψ/dt = Hψ; the sign of the hopping ($H_{n,n+1}$ = +J or −J); A = odd sites; J_1 inside the unit cell; "long" bond = weak coupling; Python + NumPy/SciPy/Matplotlib. The hopping sign changes the answer to task 1 and which dimer state is in the lower band in task 9.
- **Order:** move task 8 (k-space) before task 5. Task 6 already asks for the "bandstructure", which is only computed in task 8.
- **Workload:** 13 tasks do not fit a few 90-min sessions. Suggested plan: (1) tasks 0–2; (2) 3, 4, k-space; (3) 5–6; (4) 7; (5) 9; (6) 10–11; (7) 12.

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0) For a chain of 20 sites in real-space with $J = 1$

0a) calculate the eigenvalues and eigenstates of the real-space Hamiltonian

0b) plot eigenvalues as dots (fig. 1) and eigenstates as 2D colorplot (fig. 2)



Task 0 — add a check students can do alone: for $H_{n,n+1}$ = +J the exact result is $E_n = 2J\cos(n\pi/(N+1))$, $\psi_n(j) \propto \sin(nj\pi/(N+1))$, n = 1…N. Numerics agree to 10^{-15} ✓.

Chain figure — good: it fixes A = odd sites, J_1 inside the cell (solid), J_2 between cells (dashed). Add a caption and refer to it in task 1. The text calls the students' plots "fig. 1–4" while this picture has no number; label it differently (e.g. "Sketch 1").

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1) For an SSH chain of 20 sites in real-space with $J_2 = 1, J_1 = 2$

1a) calculate the eigenvalues and eigenstates of the real-space Hamiltonian

1b) plot eigenvalues as dots (fig. 3) and eigenstates as 2D colorplot (fig. 4)

What structural difference do you perceive in the eigenstates of the upper or lower band?

Task 1 — with J_1 = 2 on the first bond, the N = 20 chain is in the *trivial* phase: no states in the gap ✓. Fine if intended; then add "now swap: J_1 = 1, J_2 = 2" → two edge states at E = $\pm 1.5\cdot 10^{-3}$ ✓ (not exactly 0 because the two edge states overlap weakly).

Expected answer (for +J): lower-band states are antisymmetric inside each J_1 dimer ($\psi_A\psi_B < 0$), upper-band states symmetric ✓. With −J hopping it is reversed — another reason to state the sign.

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2) Change the number of sites to 21.

2a) How does the state belonging to E=0 look like?

2b) How does it change when you vary J_1?

Task 2 — 21 sites: one exact E = 0 state, living only on A (odd) sites ✓. Why: on every B site, zero energy requires $J_1 a_n + J_2 a_{n+1} = 0$, so $a_{n+1} = -(J_1/J_2)\, a_n$.

2b expected answer: $J_1 > J_2$ → state at the right end (site 21); $J_1 < J_2$ → left end (site 1) ✓; $J_1 = J_2$ → spread over the chain. Decay length $\xi = 1/|\ln(J_1/J_2)|$ unit cells. Suggest a range: "vary J_1 from 0.5 to 2 with J_2 = 1".

Wording: "How does … look like" → "What does … look like".

3) Add an on-site potential of +0.5 to the A and -0.5 to the B-sites.

3a) How does the edge state change?

Task 3 — say which chain: the edge state exists only in the N = 21 chain of task 2 (the N = 20 chain of task 1 has none).

Expected answer: the shape does not change at all (overlap 1.0000); only the energy moves from 0 to +0.5 ✓, because the state lives only on A sites, which all get +0.5. Physics: the staggered potential breaks chiral (sublattice) symmetry, so E = 0 is no longer protected.

Suggested 3b: N = 20 with $J_1 = 1$, $J_2 = 2$: the two overlapping edge states separate into left (A) at +0.5 and right (B) at −0.5 ✓.

4) Repeat task 1, but make a long-long defect in the center of the chain.

Task 4 — define "long-long": two neighbouring weak bonds (J_2).

Problem with N = 20: a domain wall in the middle forces the right half to end on a weak bond, so the defect state pairs with an edge state at the right end. You get two states at ±0.042, not a clean zero mode ✓.

Fix: N = 21 with mirrored bonds (J_1,J_2,…,J_1,J_2 | J_2,J_1,…,J_2,J_1): both ends trivial, one E = 0 state centred on site 11, only on odd sites ✓. Optional 4b: "Why does N = 20 not give a clean zero mode?"

5) Calculate the time-evolution in the SSH chain when you excite the central site (for N=41 total sites) -> Runge-Kutta

Task 5 — say whether the chain has the defect. Doing both is instructive:

- **With the centre defect** (N = 41 version of task 4): 60 % of the light stays at the centre in the defect mode ✓ ($|\psi_0(\text{centre})|^2 = 1/(1 + 2\Sigma_{n\geq 1} 4^{-n}) = 0.6$).
- **Without defect:** the light spreads into both bands. N = 41 with J_1 first also has an edge state at site 41 ✓; a centre excitation barely touches it.

Numerics: RK4 is stable only for $dt < 2\sqrt{2}/\|H\| \approx 0.94$ ($\|H\| \leq J_1+J_2 = 3$) and is not exactly unitary. Ask students to plot the norm vs. time and compare with the exact $\psi(t) = V\, e^{-iEt}\, V^{\dagger}\, \psi(0)$.

6) Plot the Fourier transform of the propagation field to see the population in the „bandstructure"

6a) Which parameters do you have to change to increase the resolution of the „bandstructure"?

6b) What happens if you change the intitial conditions (excitation spot)?

6c) Calculate the overlap of the excitation with the eigenstates of the real-space Hamiltonian.

Task 6 — the bandstructure is only computed in task 8: move task 8 before this, so students can overlay $E(k) = \pm|J_1 + J_2 e^{ik}|$. A Fourier transform over the site index gives k per *site*: two Brillouin zones of the 2-site lattice (extended-zone picture) — warn them.

6a expected answer: $\Delta k = 2\pi/N$, energy resolution $2\pi/t_{max}$, largest energy π/dt (Nyquist). Limit: the fastest light ($v_{max} = \min(J_1,J_2) = 1$ unit cell per unit time ✓) reaches the ends of N = 41 at $t \approx 10$; after that reflections distort the spectrum, so a longer t_{max} needs a larger N.

Typo: "intitial" → "initial".

7) In an SSH chain (with or without defect) modulate the couplings in the following way:

$$J_2 = 1 + A \cdot \cos\left(\frac{2\pi}{T} t\right), J_1 = 2 - A \cdot \cos\left(\frac{2\pi}{T} t\right)$$

7a) Calculate the time-evolution.

7b) Plot the Fourier transform of the propagation field to see the population in the „bandstructure".

7c) What happens when you change T? What happens when you change A?

Task 7 — give the scales: the static bands span ±[1, 3] (gap 2, total width 6). With $\omega = 2\pi/T$:

- $T \gg 3$ ($\omega \ll$ gap): slow, the bands follow the modulation;
- $T \approx 1$–3 (ω between gap and full width): resonant, bands couple, Floquet side bands appear at $E \pm n\omega$ in the spectrum;
- $T < 1$ ($\omega >$ full width): fast, the chain behaves like the time-averaged one.

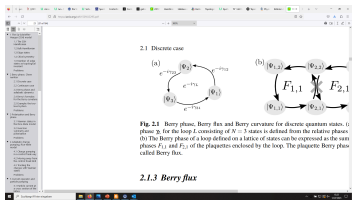
Add 7d: "What happens for A > 0.5?" — then $J_1 = J_2$ at some times, so the gap closes twice per period.

"With or without defect": say which one to do first.

8) k-space Hamiltonian for an SSH chain

8a) Calculate the eigenvalues and eigenstates of the k-space Hamiltonian for an SSH chain without defect for $k \in [0, 2\pi[$

8b) Calculate the Zak phase.



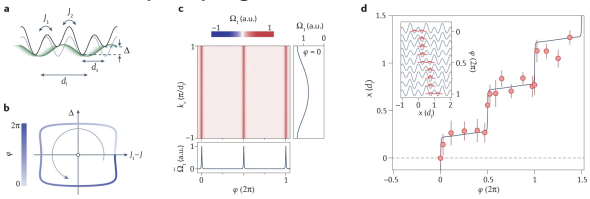
8a: with A, B in one cell, H(k) = [[0, $J_1 + J_2e^{-ik}$], [$J_1 + J_2e^{ik}$, 0]], E = ±|$J_1 + J_2e^{ik}$|. Ask students to compare with the real-space eigenvalues of task 1.

Add 8c (winding number): plot Re vs. Im of $J_1 + J_2e^{ik}$ for k ∈ [0, 2π[. It circles the origin 0 times for $J_1 > J_2$ and once for $J_1 < J_2$.

8b: numerical eigenvectors have random phases, so integrating i⟨u|∂_ku⟩ fails. Write the formula next to the figure: γ = −arg(⟨u_1|u_2⟩⟨u_2|u_3⟩…⟨u_N|u_1⟩)). Result: 0 for $J_1 = 2$, $J_2 = 1$ and π for $J_1 = 1$, $J_2 = 2$ ✓. It depends on the unit-cell choice — state it.

Figure: it is a browser screenshot (tabs, taskbar, date, cut caption). Replace by a clean crop and cite: J. K. Asbóth, L. Oroszlány, A. Pályi, *A Short Course on Topological Insulators*, Lecture Notes in Physics 919 (Springer, 2016), arXiv:1509.02295, Fig. 2.1. Its chapters 1–4 (SSH, Berry phase, polarization, Rice–Mele pumping) cover tasks 1–9: good reading for the students.

9) Thouless pumping:



Now modulate couplings and onsite potential difference in the Rice-Mele model as:

$$J_1 = 1 + \delta J \cdot \cos\left(\frac{2\pi}{T}t\right),\ J_2 = 1 - \delta J \cdot \cos\left(\frac{2\pi}{T}t\right)$$

and $\Delta = \Delta_0 \cdot \sin\left(\frac{2\pi}{T}t\right)$

with $\delta J = 0.5$ and $\Delta_0 = 0.2$, $T = 20$.

Task 9 — main problem: with δJ = 0.5, Δ_0 = 0.2, T = 20 the pumping is *not* quantized.

- **Too fast:** the smallest gap is $2\Delta_0$ = 0.4 (where $J_1 = J_2$), while the drive frequency is 2π/20 = 0.31 — not much smaller than the gap.
- **Ambiguous:** "onsite potential difference" can mean ±Δ (gap 0.4) or ±Δ/2 (gap 0.2). State it.

Simulated centre-of-mass shift after one cycle, Wannier-state start ✓:

Parameters	shift / cycle	chain needed
yours, T = 20	0.64	—
yours, T = 100	0.89	—
yours, T = 400	1.00	≈1500 cells
δJ = 1, Δ_0 = 0.5, T = 100	0.999	400 cells

Suggested parameters: δJ = 1, Δ_0 = 0.5, +Δ on A and −Δ on B, T = 100. At t = 0 then J_2 = 0: the chain is isolated dimers, so the dimer state (A−B)/√2 is an exact lower-band Wannier state — easy to launch. Both parameter sets give the staircase of two 0.5 steps per cycle ✓.

Figure: (1) no source — cite it (looks like the cold-atom pump of Lohse et al., Nat. Phys. 12, 350 (2016); please confirm). (2) There the steps are at φ = 0 and 0.5; with your formulas they are at t/T = 0.25 and 0.75, where $J_1 = J_2$ ✓. Shift the phase or tell the students. (3) The experiment uses a uniformly filled band; a single-site excitation will not reproduce panel d.

9a) For each time-step, calculate the eigenvalues and eigenstates of the k-space Hamiltonian. Plot the eigenvalues as a 3D plot along t/T and k.

9b) For each time-step, calculate the eigenvalues and eigenstates of the real-space Hamiltonian. Plot the eigenvalues as points along t/T.

9c)* Do a Runge-Kutta time-propagation and plot the center of mass over t/T. Is it quantized? Why/why not? What would you have to change to make it quantized/not quantized?

9a: fine. Also ask for the gap vs. t/T (minimum $2\Delta_0$).

9b: say "open boundaries". Then edge states cross the gap in every cycle — the real-space picture of pumping.

9c: the starting state must be specified; it decides the answer.

- **Single site:** shift ≈ 0 ✓. It fills both bands about equally; their Chern numbers are +1 and −1 and cancel.
- **Dimer (A−B)/√2**, suggested parameters: 0.889, 0.967, 0.999 unit cells per cycle for T = 20, 50, 100 ✓ — a clean "why not quantized → adiabaticity" answer.

Add 9d: Zak phase (method of 8b) vs. t/T. It winds by 2π per cycle ✓ = Chern number 1 = one unit cell pumped. This links tasks 8 and 9.

10) Repeat Task 5 using beam propagation code

10a) How do you change the couplings J_1 and J_2?

10b) How do you add on-site potential? What happens to the edge state when you add it?

Task 10 — mismatch: task 5 (centre excitation) has no edge state, but 10b asks about one. Suggest: "Repeat tasks 2, 3 and 5 with BPM; for 10b excite the edge waveguide."

10a expected answer: through the waveguide separation d (J falls roughly exponentially with d). Add a calibration step: simulate two waveguides, measure the transfer length $L_c = \pi/(2J)$, get J(d). This two-waveguide coupler is also the start of task 11.

10b expected answer: change the refractive index (Δn) or the width of one sublattice → detuned propagation constant. A width change also changes the coupling, so the edge state will not keep exactly the same shape as in task 3; Δn is cleaner.

Missing: wavelength, cladding index, Δn, width, gap, grid, step size, length — and is a BPM code provided?

11) Mach-Zehnder Interferometers and the Bloch sphere

Task 11 — only a title. Suggested content: (a) two coupled waveguides: full power transfer, find the 50:50 length; (b) map the two amplitudes (a, b) to a Bloch vector and show the coupler as a rotation on the sphere; (c) build an MZI from two 50:50 couplers and a phase φ; plot the output vs. φ (sinusoidal fringes).

The waveguide-lab app in your repo (photonics-tutorial/waveguide-lab) already has MZI phase routing and a two-mode Bloch sphere (coupled-mode, not BPM) and could be reused.

12) Coding with AI agents

Task 12 — only a title. Suggestion: students redo one earlier task with an AI agent and must verify the output against the checks above (E_n formula of task 0, Zak phase 0/π, pumped charge 1, norm conservation). Discuss typical agent mistakes: sign conventions, random eigenvector phases, RK4 step size. Rule: core code written by hand first.